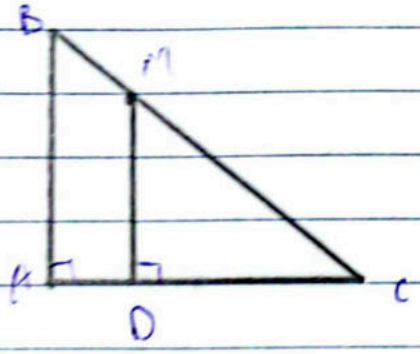
Bài 1: Văn Hải



Xét có: $AB \perp AC$

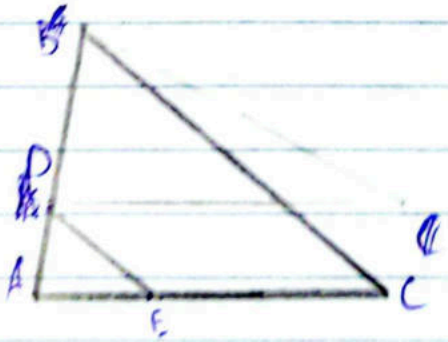
$MD \perp AC$

$\Rightarrow MD \parallel AB$

Xét $\triangle ABC$ có $MD \parallel AB$

$\Rightarrow \triangle AMD \backsim \triangle ABC$ (Thales)

Bài 2:



Có: $\left.\begin{array}{l} \frac{AD}{AB} = \frac{2}{6} = \frac{1}{3} \\ \frac{AE}{AC} = \frac{3}{9} = \frac{1}{3} \end{array}\right\}$

$\Rightarrow \frac{AD}{AB} = \frac{AE}{AC}$

Xét $\triangle ADE$ và $\triangle ABC$ có:

$\frac{AD}{AB} = \frac{AE}{AC}$ (cmt)

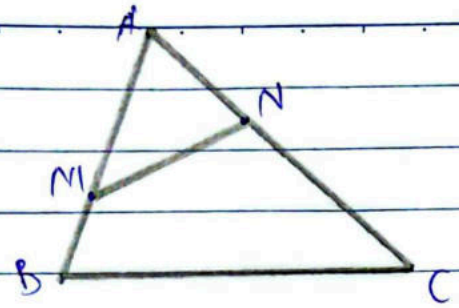
$\widehat{BAC}$ chung

$\Rightarrow \triangle ADE \backsim \triangle ABC$ (c.g.c)

$\Rightarrow \widehat{ADE} = \widehat{ABC}$

$\Rightarrow DE \parallel BC$ (2 góc đồng vị)

Được quét bằng CamScanner



Có: $\frac{AM}{AC} = \frac{4}{8} = \frac{1}{2}$

$\frac{AN}{AB} = \frac{3}{6} = \frac{1}{2}$

Xét $\triangle AMN$ và $\triangle ACB$ có:

$\frac{AM}{AC} = \frac{AN}{AB}$ $(= \frac{1}{2})$

Chung $\widehat{BAC}$

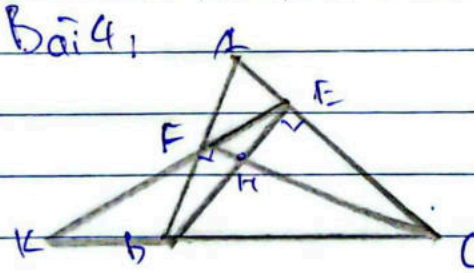
$\Rightarrow \triangle AMN \backsim \triangle ACB$ (c.g.c)

Có $\frac{AM}{AC} = \frac{AN}{AB}$

$\Rightarrow \frac{AM}{AN} = \frac{AC}{AB}$

$\Rightarrow AM \cdot AB = AN \cdot AC$

Bài 4:



a) Xét $\triangle ABE$ và $\triangle ACF$ có:

$\widehat{AFC} = \widehat{AEB} (= 90^\circ)$

Chung $\widehat{BAC}$

$\Rightarrow \triangle ABE \backsim \triangle ACF$ (g.g)

$\Rightarrow \frac{AE}{AF} = \frac{AB}{AC} \Rightarrow AF \cdot AB = AE \cdot AC$

Được quét bằng CamScanner

b) Có: $\frac{AE}{AF} = \frac{AB}{AC}$ (cmt)

$\Rightarrow \frac{AE}{AB} = \frac{AF}{AC}$

Xét $\triangle AEF$ và $\triangle ABC$ có:

$\frac{AE}{AB} = \frac{AF}{AC}$ (cmt)

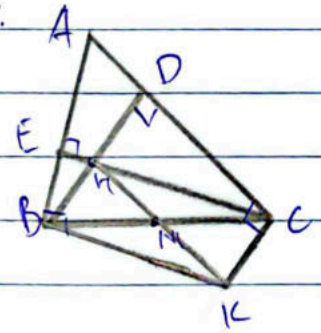
Chung $\widehat{BAC}$

$\Rightarrow \triangle AEF \backsim \triangle ABC$ (cg.c)

$\Rightarrow \widehat{AEF} = \widehat{ABC}$

c)

Bài 5:



a) Xét $\triangle ADB$ và $\triangle AEC$ có:

$\widehat{AEC} = \widehat{ADB}$ ($=90^0$)

Chung $\widehat{BAC}$

$\Rightarrow \triangle ADB \backsim \triangle AEC$ (cg.cg)

b) Có $\triangle ADB \backsim \triangle AEC$ (cmt)

$\Rightarrow \widehat{ABD} = \widehat{ACE}$

Xét $\triangle EHB$ và $\triangle DHC$ có:

$\widehat{ABD} = \widehat{ACE}$ (cmt)

$\widehat{BEH} = \widehat{CDH}$ ($=90^0$)

$\Rightarrow \triangle EHB \backsim \triangle DHC$ (cg.cg)

$\Rightarrow \frac{HE}{HD} = \frac{HB}{HC}$

$\Rightarrow \frac{HE}{HB} = \frac{HD}{HC} \Rightarrow HE.HC = HB.HD$

0 5 10 15 20 HAI TIEN

Được quét bằng CamScanner